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A Sensitivity-Aware and PSO-Driven Differential Privacy Method with Customized Budgets for Structured Data Perturbation

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***Abstract*—Differential privacy (DP) is the leading standard for privacy protection, providing rigorous privacy guarantees for various data. However, its conventional approach of treating all records uniformly regarding privacy risk and using a non-adaptive privacy budget (ϵ) often compromises data utility in subsequent analyses. This uniform treatment and fixed ϵ can introduce significant perturbations, making the secondary use of shared data challenging. To overcome these limitations, we introduce a novel record-sensitivity-aware and Particle Swarm Optimization (PSO)-driven customised ϵ-DP method for data perturbation. Our approach significantly enhances data utility without compromising privacy in data sharing by introducing three key optimisations to the traditional DP framework:**

First, we partition records into three sensitivity classes (high, medium, and low) based on the privacy risk. Second, we adopt a PSO mechanism to determine the optimal ϵ for each partition, perturbing data with a variable ϵ that considers sensitivity, rather than using a single, fixed ϵ for the entire dataset. Finally, noise is injected by grouping attributes horizontally, rather than adding noise to each attribute independently, to prevent the generation of inconsistent values in the perturbed data. Detailed experiments on real benchmark and synthetic datasets demonstrate the superiority of our method in terms of utility and privacy across seven evaluation metrics, compared to the latest state-of-the-art ϵ-DP methods.

***Index Terms*—record sensitivity, particle swarm optimization, privacy, utility, differential privacy, perturbation, noise**

I. Introduction

DATA owners such as hospitals, banks, and government agencies collect a massive amount of personal data from their clients as a part of their day-to-day operations and to improve service quality. In hospitals, collected data can be used to investigate demographic associations with certain diseases or to develop new personalized treatment methods [1]. In other sectors, data can offer many business opportunities (or new revenue models) when analyzed with the latest data mining tools. However, the existence of sensitive information in the data (e.g., diseases, finances, professions) limits its use, and data owners are obliged to use privacy preservation (PP) methods to remain legally compliant [2]. Recent regulations bind most data owners to ensure strict privacy guarantees when processing and sharing users' data, and they employ robust PP methods to prevent privacy violations [3], [4]. To this end, data undergoes modifications to prevent leakage of sensitive information while still permitting knowledge discovery. Two widely used PP solutions in data sharing (DS) are *anonymization* and *differential privacy* (DP).

The anonymization methods, i.e., k-anonymity, ℓ-diversity, and t-closeness as well as their upgraded versions, have been widely used for PP in DS [5]. However, these methods are not secure when the adversary has extensive background knowledge or when a large amount of auxiliary data is available. Besides, these methods introduce greater modifications to the data, resulting in reduced utility. In contrast, DP is a rigorous PP method with proven mathematical guarantees independent of the data content and an attacker's knowledge [6]. Due to robust privacy guarantees and higher indistinguishability, it has been preferred over anonymization for PP in DS. However, in DP, it is challenging to determine the optimal privacy budget ϵ, and most DP methods yield poor utility in downstream tasks (e.g., query answers, ML training, etc.) [7].

To address poor utility issues, DP methods have recently undergone extensive upgrading, such as combining k-anonymity with DP [8], determining a data-characteristic-aware adaptive ϵ [9], reasonable allocation of ϵ by dividing data into low-dimensional clusters [10], attribute-correlation noise addition [11], noise injection based on the importance of attributes [12], stochastic influence-aware perturbation [13], and less noise addition based on correlation analysis [14]. Despite these developments, it is still challenging to guarantee optimal utility with strong PP. Four crucial problems with prior DP-based methods are as follows.

- Existing DP methods do not provide algorithmic solutions for classifying records based on privacy risk (i.e., sensitivity), and they use a fixed ϵ while perturbing data. Ignoring record sensitivity and instead using the same ϵ for the entire dataset can significantly degrade data utility.
- Some DP methods determine a suitable ϵ but delve deep into the local structure of the data (cluster-level ϵ), and the cost function is not flexible (e.g., it favors privacy across all clusters), which cannot determine the optimal ϵ that resolves the crucial privacy-utility trade-off.
- Most DP methods add noise to each attribute, one after another (without creating combinations), which can lead to generation of inconsistent values that do not exist in the real data, thereby affecting its secondary use.
- Majority of DP-based methods have poor utility in downstream tasks (e.g., building classifiers), owing to utilizing one fixed ϵ for entire dataset and not transferring the

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sensitivity of the query function to more records/values.

Resolution of the above-cited problems is imperative because perturbed data is widely used in downstream tasks or for decision-making. Heavily perturbed data can hinder the knowledge extraction process, and can limit secondary use, yielding no dividends to data owners or users. Furthermore, some high-stakes tasks and 4P goals (*personalized, participatory, predictive, preventive*) require the development of secure DS methods for secondary data use [15].

A. Major contributions of this work

To address the challenges mentioned above, we propose a DP method for structured data perturbation by introducing three novel optimizations. The innovation in this work is to offer an algorithmic solution for data classification w.r.t. sensitivity when users cannot determine the sensitivity of their data [16], finding the optimal ϵ and utilizing a distinct privacy budget (rather than a fixed value), and by generating perturbed data that closely resembles the real data. The key idea is to classify records in the real data as low, medium, and high sensitivity based on the privacy risk, which serves as the foundation for deriving a more tailored ϵ for each partition. To find the optimal ϵ for each partition, we employ particle swarm optimization (PSO) with careful modifications in the cost function (privacy and utility requirements are directly embedded). Lastly, exponential and Laplace mechanisms are applied to the categorical and numerical parts of the real data to curate perturbed data. Specifically, we group categorical attributes to prevent inconsistent value generation, and we inject noise directly into the values of numerical attributes to augment utility. To the best of our knowledge, all three optimizations are made jointly for the first time in DP to address the privacy-verus-utility issues effectively. Our key contributions are as follows.

- We explore design- and performance-related drawbacks in recent DP-based methods that mostly rely on users to define the sensitivity of their data, or that use a fixed ϵ for the entire dataset, and we identify opportunities to propose and implement the novel record-sensitivity and PSO-driven customized ϵ-based DP method, which can effectively preserve the utility needed for downstream tasks from the perturbed data without risking user privacy.
- We introduce the record-sensitivity concept, which classifies an entire record into one of three classes to reduce noise in less sensitive records. In contrast, prior research considers only the sensitivity of some attributes in the record, offering no concrete solution to enhance utility.
- We employ a PSO algorithm with modifications, i.e., the cost function of the PSO computes the optimal ϵ for three classes of data while considering both utility and privacy as an objective function, whereas prior research only considers privacy, which cannot derive an optimal ϵ to resolve the crucial privacy-utility trade-off.
- We employ the combination concept on categorical parts of the data, where values are horizontally grouped into combinations before application of the exponential mechanism in order to keep semantics in the perturbed data as similar as possible to the real data. Laplace noise is applied directly to numerical data by transferring sensitivity to multiple values, resulting in more accurate aggregate statistics.
- Extensive experiments on both real and synthetic datasets by using seven evaluation metrics, prove that our method can yield significantly better utility without sacrificing privacy compared to existing SOTA DP-based methods.

The rest of this paper is organized as follows. Section II discusses the background material and related SOTA studies. Section III provides essential preliminary information. Section IV presents the system model of this research, along with the associated models. Section V presents our proposed method in detail. Section VI presents experimental results and comparisons of our method versus other methods on real and synthetic datasets. Finally, we conclude the paper in Section VII.

II. Background and Analysis of SOTA PP Methods

In this section, we provide background to the subject matter and analysis of SOTA PP methods used in DS.

A. Background

Structured/tabular data is often used in the privacy literature, wherein four different types of attributes are typically included: quasi-identifiers (QIDs), sensitive attributes (SAs), non-sensitive attributes (NSAs), and explicit identifiers (EIs). The definition and examples of each attribute were discussed in Arshad and Asghar [17]. Each attribute type is processed differently during the anonymization/noising process, depending on the PP method and the DS goals. There are two mainstream solutions for PP in data analysis/sharing presented in Table I. Next, we discuss some of the latest SOTA methods that are used for PP in DS.

B. Analysis of syntactic/anonymization PP methods

Syntactic/anonymization methods are widely used for PP in realistic scenarios owing to their low computing overhead and conceptual simplicity. k-Anonymity is the leading anonymization method, where the raw data is divided into equivalent classes, the size of each class is at least k, and the values under QIDs are transformed into general values to preserve privacy [18]. The disclosure risk under k-anonymity is $1/k$. However, it can lead to explicit SA disclosure if the attacker has background information about the target user and all SA values in the class/cluster are identical. Information loss from the k-anonymity model is also very high due to generalization/suppression operations.

To prevent SA disclosure, two advanced models (ℓ-diversity [19] and t-closeness [20]) were introduced that ensure heterogeneity in SA values and bind the SA distribution w.r.t. the real data. Recently, these methods have undergone extensive enhancements to better balance privacy and utility in DS. Su et al. [21] enhanced the k-anonymity to defend against two commonly encountered attacks (skewness and similarity) in the conventional k-anonymity model. Li and He [22] developed a local generalization and bucketization-based method for protecting identity and SA disclosure when the data includes semi-sensitive QIDs. Chen et al. [23] developed an anonymization model to optimize the privacy-utility trade-off by exploiting characteristics of the real data.

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Table I
COMPARISONS OF TWO WELL-KNOWN SOLUTIONS FOR PP IN BIG DATA ANALYSIS/SHARING.

Name	Privacy preservation by	Main mechanisms used	Representative methods	Barriers to adoption
Syntactic	Converting raw data into coarse form	Generalization, suppression	k-anonymity and its enhancements	Poor privacy-utility guarantees
Semantic	Adding noise to the raw data	Laplace, exponential, Gaussian	Differential privacy and its enhancements	Better privacy but poor utility

Di et al. [24] developed a distributed version of k-anonymity and ℓ-diversity that preserves privacy guarantees for large datasets. Barezzani et al. [25] devised a downstream task-aware anonymity method based on k-anonymity and ℓ-diversity concepts for DS. Srijayanthi and Sethukarasi [26] developed a clustering and feature selection-based anonymity algorithm to address the trade-off between data utility and computing efficiency. Onesimu et al. [27] developed a hybrid method based on ℓ-diverse slicing and fixed intervals to securely publish health data. Mehta and Rao [28] devised scalable versions of k-anonymity (ImSkA) and ℓ-diversity (ImSℓD) for optimizing utility-versus-efficiency trade-offs while offering robust privacy guarantees. Some methods based on ARX tools have been developed to facilitate health DS without compromising privacy guarantees [29]. Besides, some anonymization methods were designed to support a specific purpose (e.g., diversity preservation w.r.t. a variety of individuals) [30]. Though anonymization methods help accomplish DS, a defense against more robust attacks (probabilistic, correlation, etc.) cannot be accomplished, and utility from anonymized data is low, which can impact statistical reproducibility [31]. Additionally, anonymity methods alter the semantics of the data too much, resulting in substantial informational/utility loss in most cases.

C. Analysis of semantic/DP-based PP methods

Semantic/DP-based methods are widely used to ensure data confidentiality while maintaining reasonable utility. In these methods, analytical results are independent of the presence or absence of individuals in the underlying data. The two famous versions of semantic methods used for PP in the big data era are local DP, which takes only one user's data as input and generates randomized responses per user, and centralized DP, which takes all the data as input and requires the output to be indistinguishable. Li et al. [32] devised a PCA-based DP algorithm for sharing high-dimensional data with personalized PP. Cheng et al. [33] proposed a DP-based method for sharing data in a multi-party environment, ensuring both privacy and the desired data utility by employing a non-overlapping covering design and dynamic programming to reduce noise. Lu et al. [34] developed Bayesian networks combined with a DP method to resolve the privacy-utility paradox in DS with an elastic ϵ allocation strategy. Some PP methods optimize DP for sharing high-dimensional data from the perspectives of perturbation error and computing efficiency [35]. Recently, a DP model was integrated with PP while synthesizing data, i.e., curating synthetic data (SD) [36].

Sun et al. [37] devised a DP method with an optimal noise design for PP when releasing correlated data. However, the proposed method cannot effectively resolve the privacy-utility trade-off, and offers limited assistance in processing categorical data. Zhang et al. [38] developed a method based on DP and information-fusion techniques for multi-party DS. The proposed method dynamically determines noise based on the training progression to optimize utility, utilizing the global-aware discriminator regularization method for PP. Yang et al. [39] developed a DP-based frequency estimation method that utilizes sampling to achieve higher accuracy without relying on a trusted server. DP has been applied to diverse scenarios, such as the internet of medical things, to share data while flexibly balancing privacy and utility [40]. Some work explored a better use of ϵ than global sensitivity in live databases [41]. Recently, DP has been optimized to yield perturbed data that is truly representative of the real data, thereby preventing incorrect analytic results [42]. Lastly, some approaches rely on evolutionary algorithms to find the best ϵ in order to yield robust privacy guarantees [43]. We affirm the contributions of these methods, but the balance between utility and privacy is not well maintained due to either fixed ϵ use or determining ϵ by favoring only one objective (e.g., privacy alone). Also, the record-sensitivity concept was not utilized in previous work to lessen the perturbation, which can lead to the four key issues stated in Section I. Our major contributions address these deficiencies in prior SOTA DP-based methods and provide a practical approach to systematically resolving them.

III. PRELIMINARIES

In this section, we introduce the data model and definitions related to the DP model. Table II summarizes the key notations used in our work.

Table II
MAIN SYMBOLS AND BRIEF DESCRIPTIONS.

Symbol	Description
T, P_T	Raw/unperturbed dataset, perturbed dataset
X_i, $X_i^{'}$	ith user in T, ith user in P_T
m, N	Number of attributes in data, number of users in data
Q, Y	Set of quasi-identifiers in T, sensitive attribute in data
X_{nu}, X_{ca}	Numerical QIDs in X, categorical QIDs in X
$x_{u_i}^{j}$, y_{u_i}	Value of the jth QID of the ith user, SA of ith user
k, $C^{'}$, C	Cluster number, initial cluster center, cluster set
$P_(C_i^{'}, u_i)$	Proximity of the ith user to the cluster center
ϵ, Δf	Privacy budget, sensitivity of the query function
ρ_l, ρ_m, ρ_h	Low-, medium-, and high-sensitivity partitions of T

A. Structured data model

In this work, we consider structured data in which each row represents a real-world individual, and each column denotes a piece of information specific to that individual. Let $T = \{X_1, X_2, \ldots, X_N\}$ denote the collected dataset from users, where X_i denotes the entire ith record/user. We consider a scenario in which T encloses m attributes, which are represented as $A = \{a_1, a_2, \ldots, a_m\}$. We assume the label for each attribute is provided, and all attributes have distinct

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values. Attributes $m-1$ are generic, and one attribute is an SA. The attributes in T are mixed (numerical and categorical). In simple terms, T is a matrix of $N \times m$ dimensions, where N shows the # of rows, and m is the # of columns. The format of T is in Table III.

Table III
A RAW T WITH N RECORDS, AND m ATTRIBUTES.

Record	a_1	a_2	...	a_m
X_1	x_1^1	x_1^2	...	x_1^m
X_2	x_2^1	x_2^2	...	x_2^m
...	...	...	...	...
X_N	x_N^1	x_N^2	...	x_N^m

B. Differential privacy and related mechanisms

With a privacy budget, $\epsilon > 0$, that curbs the privacy-utility trade-off, we require that function f to satisfy ϵ-DP per Definition 1.

Definition. 1 *(ϵ-DP): A function f, where $f : T \rightarrow R$ satisfies ϵ-DP if $\forall$ pairs T_1 and $T_2 \in T$ differing in solely one record, and for every output subset $S \subseteq R$:*

$$Pr[f(T_1) \in S] \leq e^{\epsilon} Pr[f(T_2) \in S], \tag{1}$$

where T is the dataset, function $f : T \rightarrow R$ takes T as input, and outputs R.

The DP model is based on the sensitivity of the query function, which injects noise into the query output to protect privacy. The sensitivity (Δf) is defined as follows.

Definition. 2 *(Sensitivity): Given any function $f : T \rightarrow R^d$ that maps a raw database to the vector of real number, then for all pairs (or neighboring datasets) T_1 and $T_2 \in T$, Δf is defined as follows:*

$$\Delta f = \max_{T_1, T_2 \in T} ||f(T_1) - f(T_2)||_1, \tag{2}$$

where $||.||_1$ is the L_1 norm.

The ϵ-DP can be realized via different mechanisms. Dwork [44] proposed a Laplace mechanism that adds noise to the output of the query function per the Laplace distribution (LD). Let $Lab(b)$ denote the LD bounded by position 0 and scale parameter b; then, the probability density function of LD is Eq. 3.

$$p(g) = \frac{1}{2b} e^{(\frac{-|g|}{b})} \tag{3}$$

Definition. 3 *(Laplace mechanism): Given any function $f : T \rightarrow R^d$ having Δf as sensitivity, if the output of f satisfies Eq. 4, then f is ϵ-DP:*

$$f(T) = f(T) + \sigma \tag{4}$$

where σ denotes the noise drawn from LD at scale $\frac{\Delta f}{\epsilon}$,

$$\sigma = Lap(\frac{\Delta f}{\epsilon}) \tag{5}$$

The Laplace mechanism is suitable for X_{nu}, and the exponential mechanism is devised for X_{ca} (Definition 4).

Definition. 4 *(Exponential mechanism): Any algorithm M taking T as input and producing r as output, where $r \in Range(M)$. The function $\mu(T, r) \rightarrow R$ is a function that evaluates r. If M selects r from $Range(M)$ with a probability proportional to $e^{\frac{\epsilon\mu(T,r)}{2\delta\mu}}$, then M provides ϵ-DP.*

The above mechanisms perturb T and transform it into P_T, which is privacy-preserved and can be used in downstream tasks (e.g., histogram analysis, rule derivation).

IV. SYSTEM MODEL

A. System Model

In this work, we deal with a fundamental DS scenario involving multiple entities: users(record owners), analysts (some acting as adversaries), data publishers, and data owners, as shown in Fig. 1. Users provide their information to data owners

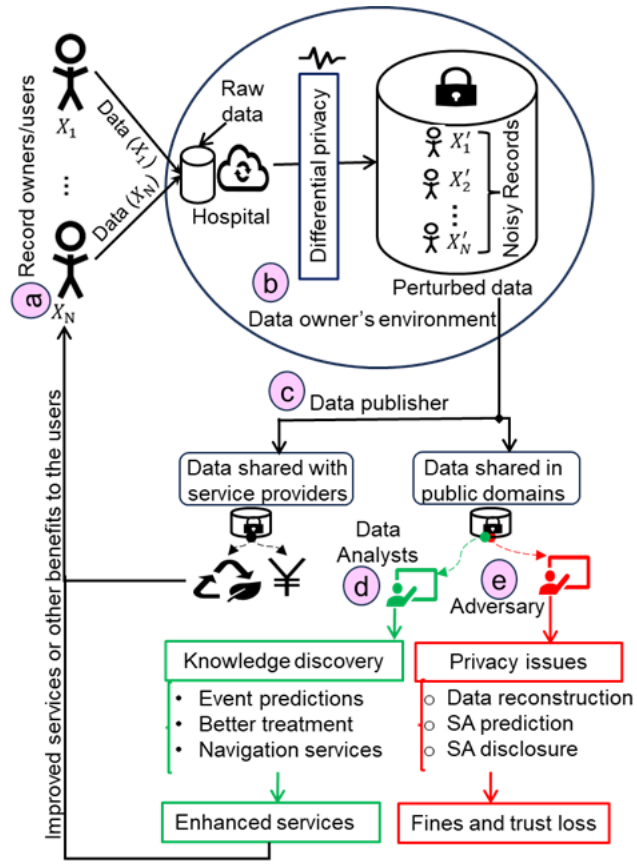

Figure 1. Schematic of the system model considered in this work. This schematic shows the five different actors involved in the data sharing process, labeled as a-e. It also shows the data conversion from raw form to perturbed form, privacy issues, and the utility of the data in real services.

in record format or tuples. The data owners are responsible for managing that data. Later, the data in perturbed form is either shared by data owners or data publishers. The released data is analyzed to extract patterns/knowledge, and the mined knowledge is employed to enhance real-life services (e.g., healthcare). Some analysts can behave as an adversary whose intention is to breach privacy in the perturbed data. In the system, each user can have different values for their attributes. Say real user u_i has information in tuples, denoted by X_i, where two types of attributes are encompassed. If N users provide data, the dataset size is $|T|$. If strict ϵ-DP is applied to T, there is a risk that true information will be distorted,

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and the discovery process can be seriously impacted. Our goal is to devise an ϵ-DP method that minimizes distortion while enhancing utility from the perturbed data, while preserving user privacy against strong adversaries.

B. The Threat Model

This work assumes that most actors involved in the system (e.g., users, database publishers, database owners) are honest, except for data analysts who may or may not be honest. Semi-honest analysts can turn into adversaries and might explore ways to threaten individual privacy. Not all types of identifiable information are present in T. Still, QIDs can be obtained from external sources and can be linked to the perturbed data to identify individuals, or SAs can be exposed. In this work, we consider more advanced attacks in terms of PP. Specifically, our method is susceptible to identity and SA leaks because adversaries might (i) accurately reconstruct the entire perturbed dataset (or targeted columns) into real form, and might compromise user privacy afterward; (ii) predict/infer SAs of target users given the QID information or by utilizing similar individuals' information from the perturbed data; and (iii) determine SAs by linking auxiliary information when QIDs are exposed as background information/knowledge, and posterior probability increases after seeing a data release. All three threats are realistic and can pose significant risks to user privacy. Therefore, we intend to protect user privacy against the three privacy threats cited above that can occur during DS.

C. Privacy model

Our perturbation method is based on DP, and two mechanisms are adopted, each with modifications, to create a perturbed version of T. Specifically, we apply the Laplace mechanism (LM) to numerical QIDs, and apply the exponential mechanism (EM) to combinations of categorical QIDs. An example of LM application to the six age values is expressed in Eq. 6, where θ is the noise added to the actual values to yield perturbed values:

$$a_1 = \begin{vmatrix} x_1^1 \\ x_2^1 \\ x_3^1 \\ x_4^1 \\ x_5^1 \\ x_6^1 \end{vmatrix} = \begin{vmatrix} 45 \\ 42 \\ 35 \\ 63 \\ 55 \\ 64 \end{vmatrix} = \begin{vmatrix} x_1^1 \pm \theta \\ x_2^1 \pm \theta \\ x_3^1 \pm \theta \\ x_4^1 \pm \theta \\ x_5^1 \pm \theta \\ x_6^1 \pm \theta \end{vmatrix} = \begin{vmatrix} 45.8 \\ 41.8 \\ 35.9 \\ 61.3 \\ 54.7 \\ 63.1 \end{vmatrix} \tag{6}$$

In Eq. 6, θ is chosen arbitrarily to illustrate the use of LM in data noising. In practice, θ is drawn from LD, and its amount can vary depending on the ϵ value. When ϵ is high, the amount of noise is small, and vice versa. The EM is applied to categorical QIDs after creating value combinations. The primary purpose of applying the EM to the combination is to prevent the generation of incorrect or nonexistent values in T_P.

D. Design goals and problem formulation

The effectiveness of our method is evaluated based on the following two key aspects of DS.

Utility-enhancement: ensures that data analysts can effectively use T_P for secondary purposes, and can extract knowledge to the maximum extent possible to improve real-world services. Specifically, it minimizes the classification error rate (CE_{rate}), query error rate (QE_{rate}), and information loss (ILoss) from T_P.

Privacy-preservation: ensures that T_P is ϵ-DP private and has higher indistinguishability compared to T. Attackers with greater background knowledge or resources cannot compromise the privacy of a target user or infer the SAs of a target user. Specifically, our method protects against data reconstruction, SA prediction/inference, and SA disclosure via linking or background knowledge.

The problem we aim to address in this work is given below.

Problem 1. *Given a person-specific dataset, T, having various X_{nu} and X_{ca} attributes (age, citizenship, race, name, gender, income/disease, etc.), how do we curate perturbed dataset T_P where (a) T_P is ϵ-DP private, (b) $SS(T \sim T_P)$—the structural similarity between T and T_P—is high, (c) T_P has good utility for analytics (e.g., histogram analysis with low ILoss, prediction/classification with the least error rate), and (d) $\forall \perp_i \in T \approx T_p$—the possibility of preserving correct combinations in T_P w.r.t. X_{ca}—is the same as T (in other words, there is no incorrect combination of categorical QIDs in T_p).*

V. The Proposed Data Perturbation Method

In this section, we discuss the proposed record-sensitivity-aware, modified PSO-driven, customized ϵ-based DP for structured data perturbation. Fig. 2 shows the workflow of the proposed method. There are five key steps that transform $T \rightarrow T_P$ while fulfilling the stated design goals.

A. Step 1: Data Pre-processing

In the first step, T is pre-processed to yield information analysis in downstream tasks. Specifically, five different operations are applied to T to give it a suitable structure.

1) Identification of attribute types present in T: As discussed earlier, four main types of attributes can be present in any real-world T. At the outset, the types of attributes in T are identified. Identifying attribute types is vital to ensure distinct treatment during the perturbation process. For example, SAs are released as is for analytical purposes, but noise is added to QIDs to preserve privacy. In this work, we first identify the types of attributes and label them as QIDs, SAs, NSAs, and EIs. The attributes can be identified by comparing each column's label with established information and taxonomies in the privacy field. For instance, age is a critical demographic factor and is therefore classified as a QID. In contrast, names, email addresses, and phone numbers are EIs because they can directly identify someone. In some cases, some types of attributes may or may not be in the underlying T. After determining the attribute types, further pre-processing is applied to each one.

2) Removal of NSA and EIs: EIs are risky in terms of privacy breaches and are removed from T as a standard statistical disclosure control (SDC) practice in the privacy literature. On the other hand, NSAs are usually not collected

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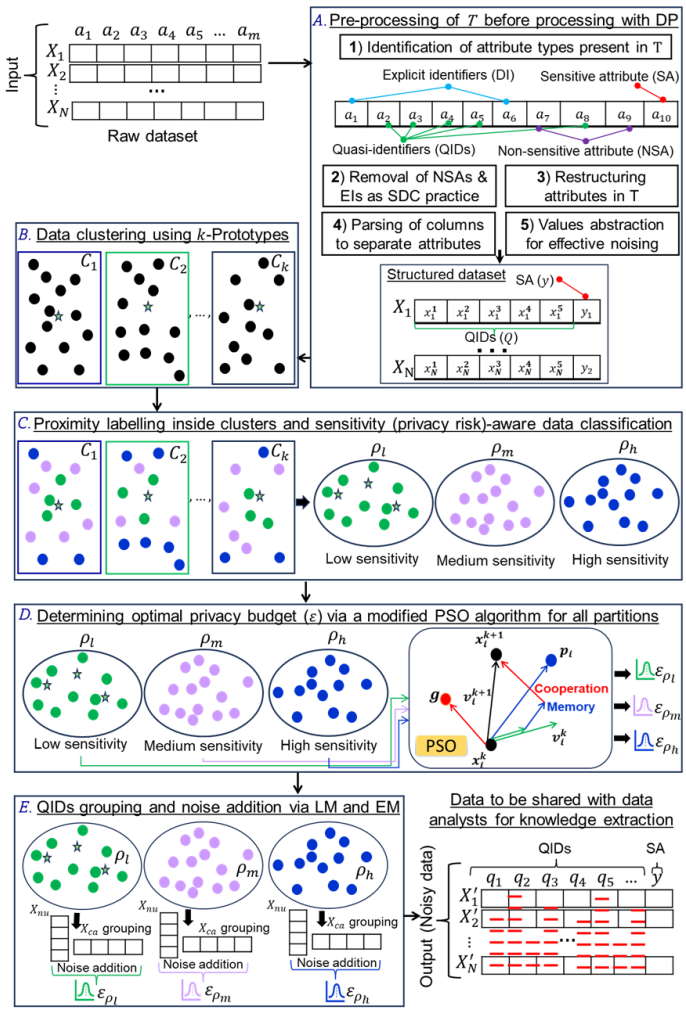

Figure 2. Workflow of our proposed data perturbation method. This schematic visualizes the five main steps (labeled as *A.*-*E.*) of the proposed method that transform T into T_P while effectively resolving the privacy-utility trade-off.

from individuals because of their limited contribution to data utility, or they are discarded before the perturbation process to conserve computing resources. In this work, we remove both types of attributes from T to reduce privacy risk and redundant computing operations. EI and NSA removals in Fig. 2 are expressed in Eq. 7 and Eq. 8:

$$T - EIs = \{a_1, a_2, a_3, a_4, a_5, a_6, \ldots, a_{10}\} - \{a_1, a_6\} \quad (7)$$

$$T - NSA = \{a_2, a_3, a_4, a_5, a_7, a_8, a_9, a_{10}\} - \{a_7, a_9\} \quad (8)$$

After removing EIs and NSAs, T preserves only QIDs and SA, as shown below.

$$T = \{a_2, a_3, a_4, a_5, a_8, a_{10}\} \quad (9)$$

In Eq. 9, $\{a_2, a_3, a_4, a_5, a_8\} \in Q$ and $a_{10} \in Y$ are QIDs and SA, respectively. From here on, most computation is performed on both attribute types because they are crucial for downstream tasks.

3) Arrangement/restructuring of attributes: EIs and NSAs can be removed from different indexes, and the structure of the remaining data can become ambiguous. To resolve this problem, we restructure attributes in T. In Fig. 2, QIDs are grouped to perform subsequent operations, whereas the SA is moved to the last column in T. In addition, we further transform the QID structure by separating them by type (i.e., X_{nu} and X_{ca}) to apply DP mechanisms in the final steps. After that, T has QIDs and the SA, where the QIDs are classified as segments of X_{nu} and X_{ca}.

4) Parsing columns: In almost every T, information about individuals/users is well classified: each row represents an individual, whereas each column represents one piece of information concerning the individual. However, sometimes the information in T is not well organized, so parsing of columns is imperative. Fig. 3(a) is an example of T with unorganized information (e.g., all the information is in one cell), which cannot be directly processed in the model. To

(a) Given data

	Age; "job", "marital"; "education"; "balance"; "housing"; "y"
1	58; "management", "married"; "tertiary"; 2143; "yes"; "no"
2	44; "technician", "single"; "secondary"; 29; "yes"; "yes"
…	…

(b) Output of parser 1

	Age	Job	Marital	Education	Balance	Housing	Y
1	58	"management"	"married"	"tertiary"	2143	"yes"	"no"
2	44	"technician"	"single"	"secondary"	29	"yes"	"yes"
…	…						

(c) Output of parser 2

	Age	Job	Marital	Education	Balance	Housing	Y
1	58	management	married	tertiary	2143	yes	no
2	44	technician	single	secondary	29	yes	yes
…	…						

Figure 3. Example of unorganized information in a T that require parsing. This figure shows data quality issues in real-world datasets and the need for proper data preprocessing before feeding them into perturbation pipelines.

resolve this issue, we developed the two parsers shown in Figs. 3(b) and (c). The first parser changes index information and allocates a unique index to each column, whereas the second parser removes commas from categorical values. The implementation of these parsers facilitates the arrangement of information in T for subsequent steps.

5) Data abstraction/transformation: Sometimes, information in a real-world T cannot be processed directly in an underlying model. For example, QIDs like a birth date can contain special characters that cannot be noised with a DP. To resolve such issues, we implement a parser that abstracts/transforms a birth date to the year alone, rather than month/day/year. With this transformation/abstraction, T becomes more organized, and computation (e.g., distance, noise) can be performed efficiently. With the help of the five pre-processing steps cited above, a well-structured T is obtained for further processing.

B. Step 2: Data clustering by using k-prototypes method

In the second step, T is divided into non-overlapping clusters using k-prototypes, which is superior to other clustering methods such as k-means. In this work, we consider mixed data (numerical and categorical), and therefore, k-prototypes is the most suitable method for clustering [45]. The k-prototypes method holds properties of both k-means and k-mode clustering via a combined dissimilarity measure. To find

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the optimal number of clusters for any T, we implement the elbow method, as shown in Fig. 4. In this example, $k = 5$ can be used to partition T into clusters.

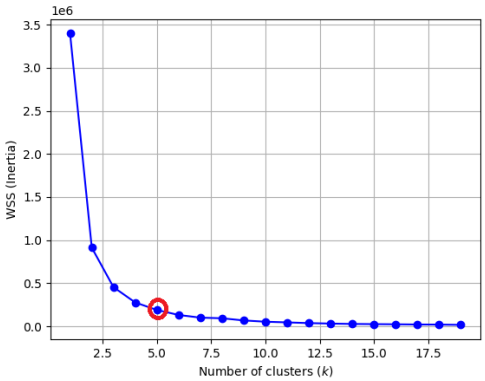

Figure 4. Example of elbow method used in finding optimal # of clusters.

After finding the k value, we divide T into clusters of a reasonable size. The objective function of k-prototypes is Eq. 10, where the goal is to find k clusters while minimizing the objective function:

$$E = \sum_{l=1}^{k} \sum_{i=1}^{N} g_{il} d(X_i, C_l^{'}) \tag{10}$$

where $C_l^{'}$ is the center of cluster C_l, g_{il} is the membership degree of record X_i belonging to cluster C_l, and $d(X_i, C_l^{'})$ is the measure of dissimilarity between X_i and $C_l^{'}$:

$$d(X_i, C_l^{'}) = \sum_{i=1}^{|X_{nu}|} (X_{ij}^{r} - w_{lj}^{r})^2 + \gamma_l \sum_{i=1}^{|X_{ca}|} \delta(X_{ij}^{c}, w_{lj}^{c}) \tag{11}$$

where X_{ij}^{r} and X_{ij}^{c} are the numerical and categorical QID values for individual i, w_{lj}^{r} is the jth numerical QID's mean in cluster l, w_{lj}^{c} is the jth categorical QID's mode in cluster l, γ_l is the weight of categorical QIDs in cluster l, and δ is the function of categorical QIDs, which is Eq. 12.

$$\delta(X_{ij}^{c}, w_{lj}^{c}) = \begin{cases} 1, X_{ij}^{c} \neq w_{lj}^{c} \\ 0, X_{ij}^{c} = w_{lj}^{c} \end{cases} \tag{12}$$

The main idea is to compute the distance between the center/prototype and the records, and the method converges when the updated prototypes/centers coincide with those from the previous iteration. The pseudocode of the algorithm that divides T into k clusters is Algorithm 1.

In Algorithm 1, T with N records and k-prototypes algorithm's parameters are the input, and a set of clusters, C, and the distance matrix (DM) of each cluster are obtained as output. There are four critical steps in the k-prototypes method: initialization, assignment, update, and convergence. After the number of clusters is determined using the elbow method, the initial cluster centers are then established. Afterward, the distance between each prototype and its records is computed, and each record is assigned to the corresponding prototype based on the distance value. Later, the prototypes are updated by using the mean and the mode information for X_{nu} and X_{ca}, respectively. The convergence check ensures the optimal number of clusters has been generated. Lastly, C and DM for each cluster are retrieved as output for further processing.

Algorithm 1 Clustering T using the k-prototypes algorithm

Require: Data set T, where $T = (X_{nu}, X_{ca})$ with N records, $X_{nu} \in \mathbb{R}^{N \times |X_{nu}|}$ (numerical), $X_{ca} \in \mathbb{R}^{N \times |X_{ca}|}$ (categorical), γ (pre-computed, set to *None*), and $init = Cao$ (density-based heuristic), max_iter (# of iterations), tol.
Ensure: C, where $C = \{C_1, ..., C_k\}$, DM (distance matrix)

```
Compute the optimal # of clusters k via the elbow method
# Initialization step
    Select k distinct records at random; assign them to C'
(initial cluster center/prototype with X_nu and X_ca).
    Initial cost Q ← ∞.
while not converged do
# Assignment step
    for i ← 1 to N do
        for j ← 1 to k do
            Compute d_ij ← X_nu + γ X_ca via Eq. 11.
        end for
        C_i ← arg min_j d_ij    ▷ assign record i to cluster j
    end for
# Update step
    for j ← 1 to k do
        if cluster C_j = ∅ then
            Re-initialize cluster center for C_j
        else
            for l ← 1 to |X_nu| do
                C^r_jl ← mean{X^r_il | C_i = j}
            end for
            for p ← 1 to |X_ca| do
                C^c_jp ← mode{X^c_ip | C_i = j}
            end for
        end if
    end for
# Convergence check
    Q_new ← Σ_{i=1}^{N} d_{C_i,i}
    if |Q − Q_new| / Q < tol then
        break
    else
        Q ← Q_new
        Continue until max_iter
    end if
end while
Compute distances of each X to its cluster centroid and add to DM
return C (Cluster set), DM (Distance matrix of clusters)
```

C. Step 3: Proximity labeling and sensitivity-aware T partitioning

After getting a set of clusters and the distance information, we process each cluster and assign a proximity label to each record. The assignment of proximity labels lays the groundwork for data partitioning based on sensitivity (i.e.,

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the privacy risk). The main idea is to analyze the distance of each record from its respective cluster center/prototype and to assign it one of three proximity labels (very close, not very close, and far away). Fig. 5 demonstrates an example of proximity-label assignment to records in a cluster.

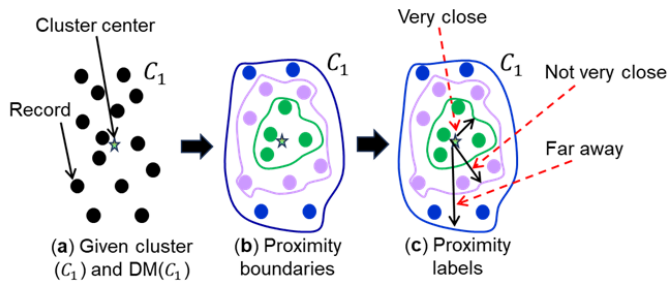

Figure 5. Example of proximity label assignment to records in cluster C_1. This schematic highlights the process for classifying records into one of the sensitivity classes using distance information between records and cluster centers.

In this example, records within the green boundary are assigned *very close* as the proximity label, records inside the purple boundary are assigned *not very close*, and records inside the blue boundary are assigned *far away*. To assign labels and draw proximity boundaries, we employ the binning concept, which is identical to the quantile concept in statistics. Specifically, we implement a quantile-based discretization function (*qcut*) to analyze the proximity between the center and the records and assign the proximity label. The pseudocode for assigning proximity labels is Algorithm 2.

Algorithm 2 Records_proximity_labelling in C

Require: Cluster set C
Ensure: $C^{''}$, where $C = \{C_1^{''}, ..., C_k^{''}\}$ (Updated clusters)

$C^{''} \leftarrow \emptyset$.
for $i \leftarrow 1$ **to** k **do**
  $\Delta_1, \Delta_2 \leftarrow C_i['DM']$, quantile $[0.33, 0.66]$
  $Bins \leftarrow [0, \Delta_1, \Delta_2, 1]$
  $P_{label} \leftarrow$ ['Very close', 'Not very close', 'Far away']
  for $j \leftarrow 1$ **to** $|C_i|$ **do**
    Analyze distance and assign one P_{label} to each X_j.
  end for
  $C_i \leftarrow assignP_{label}$ ▷ Label assigned to each record
  $C_i^{''} \leftarrow C_i^{''} \cup C_i$
end for
$C_i^{''} \leftarrow C_i^{''} \cup \{C_1, C_2, \ldots, C_k\}$
return $C^{''}$ (Cluster set with proximity label assigned)

In Algorithm 2, set of clusters C is the input, and $C^{''}$ (the updated cluster set with proximity labels assigned) is returned as output. For each cluster, we compute the quantile information and create distance bins. Subsequently, each record is analyzed with respect to the cluster center/prototype, and a proximity label P_{label} is assigned to each record. Records where the distance is $0 \sim \Delta_1$ are labeled *Very close*; records at $\Delta_1 \sim \Delta_2$ are labeled *Not very close*; records at $\Delta_2 \sim 1$ are labeled *Far away*. Ultimately, an updated cluster set is returned, with each record in each cluster having a corresponding P_{label}.

After assigning a label to each record, we create three sensitivity-aware partitions of the entire dataset. Specifically, records with identical labels in clusters are mapped to their respective classes. The pseudocode to partition the clustered data is Algorithm 3.

Algorithm 3 Sensitivity_Aware_Partitioning of $C^{''}$

Require: Cluster set $C^{''}$ where each record has a P_{label}
Ensure: ρ_l, ρ_m, ρ_h (low, medium, and high sensitivity)

$\rho_l, \rho_m, \rho_h \leftarrow \emptyset$.
for $i \leftarrow 1$ **to** k **do**
  for $j \leftarrow 1$ **to** $|C_i^{''}|$ **do**
    if ($P_{label}(X_j)$== 'Very close') **then**
      $\rho_l \leftarrow \rho_l \cup \{X_j\}$
    else if ($P_{label}(X_j)$== 'Not very close') **then**
      $\rho_m \leftarrow \rho_m \cup \{X_j\}$
    else if ($P_{label}(X_j)$== 'Far away') **then**
      $\rho_h \leftarrow \rho_h \cup \{X_j\}$
    end if
  end for
  $C_i^{''} \leftarrow \emptyset$ ▷ All records assigned to partitions
end for
return ρ_l, ρ_m, ρ_h (Sensitivity-aware partitions)

In Algorithm 3, cluster set $C^{''}$ (with each record assigned a P_{label}) is the input, and the three sensitivity-aware partitions (ρ_l, ρ_m, ρ_h) are returned as output. In each cluster, the label column is compared with each P_{label}, and the sensitivity class is determined accordingly. Finally, three partitions are returned as output for further processing. Next, we define record sensitivity in relation to privacy risk.

Definition. 5 *(Record sensitivity): Record X_i is regarded as sensitive when it has more unique values for most attributes compared to other records in T. X_i with more unique values represents specific sample-level information (a.k.a. the privacy intrusion pattern) rather than population/group-level information, which is prone to easier signaling (or re-identification).*

In [46], a precise definition of privacy was established, stating that population-level information (general patterns) is not privacy-endangering. In contrast, sample-level information (specific patterns) poses a privacy risk. Based on this definition, population-level details should be released unchanged or with less noise injected than sample-level information. Another study demonstrated that QIDs with distinct values are more identity-revealing than those with few distinct values due to higher significance values [47]. We extend these notions to the entire record and classify records into three classes based on sensitivity using shared QID space information, as shown in Fig. 6.

In Fig. 6(a), most of the records share the QID space, differing in only a few QIDs, indicating population-level information (i.e., less risky in terms of privacy disclosure). In contrast, records in Fig. 6(b) share the QID space somewhere between population and specific group information; these records are classified as medium risk. The records in Fig. 6(c) have the fewest commonalities, indicating a higher privacy risk. Any attempt to link target record X_i from this partition

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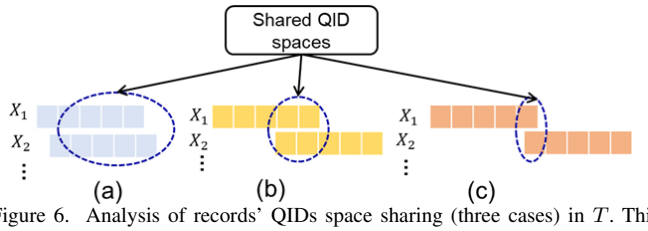

Figure 6. Analysis of records' QIDs space sharing (three cases) in T. This schematic shows the composition of QID values in T, where each record shares most, few, or too few values with other records.

results in fewer mappings, making signaling very easy (i.e., providing the least privacy preservation).

Based on the above strategy, the records are classified into three classes (e.g., low, medium, and high sensitivity). This paves the way for determining a suitable ϵ for each class, an approach that has remained unexplored in previous research.

The intuition behind the three classes in our work is to retain greater utility in T_P for data-driven services/products, because not all records/QIDs are sensitive in practice [23]. In a recent work [16], when users are empowered to specify the sensitivity of their data, they can specify sensitivity into one of the following cases: (i) none of the data is sensitive, (ii) all of their data is sensitive, and (iii) some parts of the data are sensitive while some are not. So, if we calculate the sensitivity score ρ_{score} for all three cases, it will be $|Q|$, 0, and between 0 and $|Q|$. All three cases, with sample data for three users and sensitivity scores, are shown in Fig. 7. Referring to Fig.7, there are three sensitivity cases, each requiring a different level of privacy protection: more privacy protection, moderate privacy protection, or less privacy protection. Our method is tailored to this realistic setting via three partitions, ensuring desired privacy protection by determining a customized ϵ for each partition via evolutionary optimization.

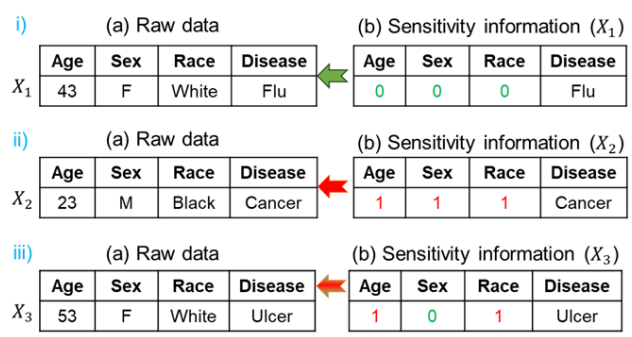

Figure 7. Example of sensitivity information (three cases) given by three users $X_1 - X_3$ for their data processing. 1 indicates that QID is sensitive, while 0 indicates that it is not.

In this work, we assume that users cannot decide the sensitivity of their data or can enter a poisoned sensitivity (regarded as an improper setting in [23]), leading to privacy leakages. To address these problems, we devise an algorithmic solution to partition data based on sensitivities without relying on users, followed by an optimized ϵ for each partition to provide desired privacy protection. It is worth noting that our method applies to both settings (with user-provided sensitivity and without) with only slight modifications. Based on the above analysis, it is fair to say that the choice of three partitions resembles real-world cases and can effectively optimize utility while safeguarding privacy. It is worth noting that some recent works partition the data into five, ten, or more clusters depending on data size [42]. However, these methods do not account for record sensitivity and cannot be applied to real-world settings with different privacy requirements. Further, in these methods, ϵ is consistently small because privacy is the sole objective in evolutionary optimization. These methods provide privacy guarantees, but they have two key problems: the privacy-versus-utility trade-off is not achieved, and the resulting T_P is not valid for data-driven services or personalized products due to higher perturbations.

D. Step 4: Determining the optimal ϵ via the modified PSO

After deriving the three partitions, we determine the optimal ϵ for each one using a modified PSO algorithm. PSO belongs to the category of swarm intelligence algorithms and is an ideal choice for optimization problems, such as parameter tuning and system optimization. Although PSO was adopted for similar purposes in [43] (e.g., finding ϵ), the objective function only considers privacy, resulting in poor utility. In this work, we build upon those findings and tailor the ϵ computation process to strike a balance between utility and privacy rather than prioritizing privacy. In previous attempts, the computation of ϵ involved analyzing the local structure of T (i.e., the clusters) without accounting for sensitivity, leading to consistently small ϵ values that significantly degrade utility. In [43], ϵ is computed for each cluster. In contrast, our method determines a suitable ϵ for all three partitions—$|\rho| >> C$ (partition size is sufficiently larger than the cluster)—by also considering sensitivity and thereby finding a more optimal ϵ. Table IV contrasts our work and existing work in terms of computing ϵ using the PSO algorithm.

Table IV
CONTRAST BETWEEN PSO USE IN PRIOR AND IN THE PROPOSED METHOD.

Criterion	OCDP (existing)	Our method
Record sensitivity	Not considered	Considered
ϵ computation	All clusters	Only three partitions
Objective function	Considers privacy	Considers privacy, utility, or both
Privacy-utility trade-off	Not solved	Effectively solved
Computed ϵ's nature	Always lower	Adaptive (e.g., higher or lower)

In Table IV, observe that the modified PSO in our method has several advantages over naive PSO implementation in prior work. Specifically, we employ three objective functions that consider sensitivity, rather than a single objective function as in existing work. Using three different objective functions, an optimal ϵ is determined for each partition, yielding better performance. Most steps in our modified PSO are the same as those in [43]. However, the objective functions differ, the three of them—privacy-only, utility-only, and both (privacy + utility)—find the optimal ϵ for each partition. The objective function to find the optimal ϵ for ρ_l is Eq. 13:

$$\mathcal{L}_{utility}(\epsilon, \alpha) = \frac{1}{|X_{nu}|} \sum_{i=1}^{|X_{nu}|} \mathbb{E}\left[\left|\tilde{\zeta}_i - \zeta_i\right|\right] + \frac{1}{|X_{ca}|} \sum_{j=1}^{|X_{ca}|} \frac{1}{t_j} \sum_{t=1}^{t_j} \mathbb{E}\left[\left|\tilde{f}_{jt} - f_{jt}\right|\right] \tag{13}$$

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where $\tilde{\zeta}_i$ and ζ_i are the noisy mean and the true mean of the ith QID; $\tilde{f}_{jt}$ and f_{jt} are the noisy and the true category, t, in the jth categorical QID; t_j denotes the total number of categories in QID j; and α is the ϵ splitting factor between X_{nu} and X_{ca}. $|X_{nu}|$ and $|X_{ca}|$ are the total numbers of numerical and categorical QIDs.

The nonlinear privacy penalty function used to find the optimal ϵ for ρ_h is Eq. 14:

$$\mathcal{L}_{\text{privacy}}(\epsilon) = \exp(\epsilon) - 1 \tag{14}$$

In Eq. 14, the function denotes the exponential privacy cost used to strongly penalize high values of ϵ (i.e., weak privacy). The objective function to find the optimal ϵ for ρ_m is Eq. 15:

$$\mathcal{L}_{\text{privacy-utility}}(\epsilon, \alpha) = \underbrace{\mathcal{L}_{\text{utility}}(\epsilon, \alpha)}_{\text{Noise-induced data error}} + \underbrace{\mathcal{L}_{\text{privacy}}(\epsilon)}_{\text{Nonlinear privacy penalty}} \tag{15}$$

where $\mathcal{L}_{\text{utility}}(\epsilon, \alpha)$ is the expected error from DP noise, and $\mathcal{L}_{\text{privacy}}(\epsilon)$ is the privacy risk cost that grows non-linearly. So, the optimization functions for all three cases are as follows:

$$(\epsilon^*, \alpha^*) = \underset{\epsilon \in [\epsilon_{\min}, \epsilon_{\max}],\ \alpha \in [0.1, 0.9]}{\arg\min} [\mathcal{L}_{\text{utility}}(\epsilon, \alpha)] \tag{16}$$

$$\epsilon^* = \underset{\epsilon \in [\epsilon_{\min}, \epsilon_{\max}]}{\arg\min} [\exp(\epsilon) - 1] \tag{17}$$

$$(\epsilon^*, \alpha^*) = \underset{\epsilon \in [\epsilon_{\min}, \epsilon_{\max}],\ \alpha \in [0.1, 0.9]}{\arg\min} [\mathcal{L}_{\text{utility}}(\epsilon, \alpha) + \mathcal{L}_{\text{privacy}}(\epsilon)] \tag{18}$$

For each partition, particle behaviors differ to find the ϵ parameter. In ρ_l, ϵ is pushed higher in order to lower the DP noise. In ρ_h, ϵ is pushed lower to increase the DP noise for better PP. In ρ_m, the optimal value for ϵ balances the privacy-utility trade-off. The pseudocode used in finding the optimal ϵ is Algorithm 4.

Algorithm 4 Compute_Optimal_ϵ for ρ_h, ρ_m, ρ_l

Require: ρ_h, ρ_m, ρ_l, and PSO parameters
Ensure: $\epsilon_{\rho_h}, \epsilon_{\rho_m}, \epsilon_{\rho_l}$

```
    ε_ρh, ε_ρm, ε_ρl ← ∅.
for i ← 1 to 3 do
    if (ρ_i == ρ_l) then
        ↓ L_utility(ε) ← via Eq. 16, and find optimal ε
        ε_ρl ← ε_ρl ∪ ε
    else if (ρ_i == ρ_h) then
        ↓ L_privacy(ε) ← via Eq. 17, and find optimal ε
        ε_ρh ← ε_ρh ∪ ε
    else if (ρ_i == ρ_m) then
        ↓ L_privacy-utility(ε) ← via Eq. 18, and find optimal ε
        ε_ρm ← ε_ρm ∪ ε
    end if
end for
return ε_ρh, ε_ρm, ε_ρl (Optimal ε for each partition)
```

In Algorithm 4, ρ_h, ρ_m, ρ_l and PSO parameters are the input, and $(\epsilon_{\rho_h}, \epsilon_{\rho_m}, \epsilon_{\rho_l})$ (the optimal ϵ for each partition) are returned as output. In this algorithm, the objective function is minimized while accounting for sensitivity to privacy risks, and optimal values for noise injection are obtained. The PSO parameters (swarm size, dimensionality, etc.) are adjusted considering the proportion of records in each partition. Through experiments, we found that Eq. 19 holds, which proves that ϵ-computing respects the privacy-utility characteristics in each partition:

$$\epsilon_{\rho_h} < \epsilon_{\rho_m} < \epsilon_{\rho_l} \tag{19}$$

where $\epsilon_{\rho_h}, \epsilon_{\rho_m}$, and ϵ_{ρ_l} are the privacy budgets for high-, medium-, and low-sensitive partitions, respectively.

E. Step 5: Data perturbations via Laplace and exponential mechanisms

In the last step, the real data in each partition is perturbed according to each ϵ computed in the previous step. Specifically, the LM is applied to the numerical data, while categorical data is perturbed via the EM. Before using the EM, we group the categorical QIDs as combinations to prevent inconsistent values from being generated in T_P. Fig. 8 shows the process of data perturbations using the variable ϵ.

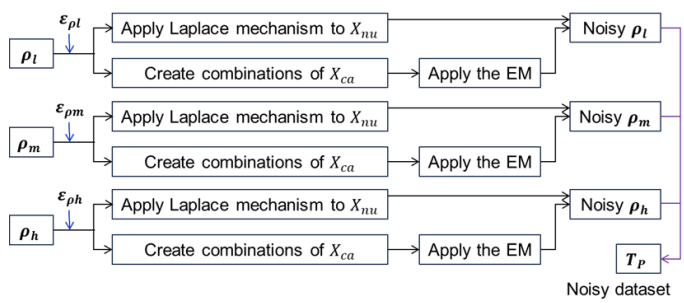

Figure 8. The data perturbation process via the EM and the LM to yield T_P.

As shown in Fig. 8, data in each partition is noised with the respective ϵ, and the final perturbed dataset T_P is obtained by combining the results of each partition. Although some methods apply the combination/grouping concept to categorical QIDs to curate truthful data [42], the perturbed numerical data can still be inconsistent due to the two operations (first replacing actual values with the cluster center, then adding noise via the LM). The application of these operations can lead to significant changes in numerical QIDs, particularly in the absence of an optimal ϵ, thereby substantially impacting analytical results. We apply the LM directly to numerical QIDs, using an optimal ϵ, resulting in only the necessary perturbations. T_P from our method can aid the knowledge discovery process, and effectively contributes to the development of data-driven services and products. Detailed privacy analysis of the proposed method is in **Appendix A**.

In real-world data-sharing contexts, the sensitivity partitioning process can be automated in the following three ways.

1) Allowing users (a.k.a. record owners) to specify the sensitivity of each item of their data, which can serve as a basis for sensitivity partitioning, as done in one of the recent works [16]. In this case, each user can provide their data in a tuple form, along with a sensitivity vector,

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where 0 and 1 indicate the sensitivity of their data, as shown in Fig. 7.

2) Employing the proposed method (i.e., applying clustering concept and performing proximity analysis between cluster center and records to partition data into three sensitivity classes), without taking sensitivity information directly from the users. This method is useful when users either cannot decide the sensitivity of their data or may specify a poisoned/wrong sensitivity.
3) Employing the hybrid method, i.e., taking sensitivity information from some users who can correctly decide the sensitivity of their data and estimating the sensitivity for the rest of the users via the proposed method.

Novel aspects of the proposed method: Our work is novel in the following ways compared to counterpart methods.

1) *Automated method for sensitivity partitioning:* This work offers a pioneering approach to partition records based on sensitivity to not waste privacy on less sensitive records, whereas existing work mostly treat each record is equally sensitive [6], [10], [12]–[14], offering limited/no support for personalized privacy needs.
2) *Three objective functions in evolutionary optimization:* To the best of the authors' knowledge, this is the first work that introduces three objective functions in evolutionary optimization (e.g., PSO) to meet personalized privacy needs. In contrast, existing methods have employed a single objective function [43], which cannot effectively resolve the privacy-utility trade-off because it introduces additional noise.
3) *Minimal and necessary data perturbation:*. This work introduces only minimal, necessary perturbations in each partition, where each partition is larger than the cluster size. In contrast, the existing cluster-DP method introduces heavier perturbations in the final data, either due to a unified or a small ϵ (e.g., privacy budget) [42], [43].

In addition to the above novel aspects, our method was tested on both real and synthetic datasets to prove its universality in modern data-sharing scenarios. Lastly, our method was evaluated using more matrices than in existing works.

VI. Performance evaluation

In this section, we evaluate the performance of the proposed method and compare its results with those of prior SOTA DP-based methods. The following subsections provide details on datasets, the implementation setup, baselines, metrics, empirical results, and comparisons.

A. Datasets

We used two real-world benchmark datasets (Bank Marketing and COMPAS) and one synthetic dataset (Adults) to evaluate and contrast the performance of the proposed method. All datasets are available at the UCI ML repository[1]. A concise description of each T is in Table V.

Bank Marketing is a high-dimensional dataset curated via direct marketing campaigns (phone calls) by a Portuguese banking institution. The goal was to determine whether the client would subscribe to a term deposit (variable y) based on demographics. The COMPAS dataset has various demographics, and the SA is the likelihood of a criminal re-offending. The Adults dataset is census information, and the goal was to determine whether a person's income was over/under $50K a year. All datasets contain mixed QIDs in their records (both numerical and categorical). All datasets were pre-processed before the experiments, as described in Section V. We used highly relevant and mixed QIDs from each T, as done in prior studies. We used ten, five, and nine QIDs from the Bank Marketing, COMPAS, and Adults (Synthetic) datasets, respectively.

Table V
Details of each T used in the experiment evaluations.

T	$\|T\|$	$\|X_{nu}\|$	$\|X_{ca}\|$	Dimensionality	SA label
Bank Marketing	45,211	6	10	16	Term deposit (y)
COMPAS	11,578	5	8	13	c_charge_degree
Adults (Synthetic)	32,561	6	9	15	Salary

B. Implementation setting

The hardware used for performance evaluation was an AMD Ryzen 9 (7000 series) notebook with an NVIDIA GeForce 4060 GPU and 16 GB of RAM. Implementation of the concept was under Python 3.9.13 with the help of multiple libraries, including *scipy, kmodes, Scikit Learn, Pandas, NumPy, itertools, pyswarm,* and *collections*. There are five key implementation steps as discussed in Section IV. We created a synthetic version of the Adults dataset for experimental purposes by using the CTGAN model[2]. The synthetic data generation process was guided by a condition to evenly explore all segments of the real data during training, thereby yielding high-quality, diverse SD. The essential parameters we use in k-prototypes are $max_iter = 50$, $tol = 0.0001 \pm \Theta$, $k = |C|$ where k is distinct for each dataset. We used pre-computed/default values for γ and $init$. We created seven, five, and ten clusters from the Adults, COMPAS, and Bank Marketing datasets, respectively. The main parameters used in our PSO are in Table VI.

Table VI
Parameters of the PSO algorithm used in computing ϵ.

Parameter name	Value	Parameter name	Value
Number of particles in the swarm	30	Number of iterations	70
Number of dimensions	1,2	Weight for inertia	0.89
Weight for a particle's own best position	0.5	Entire swarm's best weight	0.29
Number of fitness functions	3	Optimization criteria	↓ (minimize)

Apart from the values listed in Table VI, we let the PSO algorithm find the best values for some parameters (e.g., α). With the help of the above parameters and selection strategies, we achieved well-separated clusters, sensitivity-aware partitions, and optimal ϵ values. We generated multiple perturbed data sets to contrast our method with its counterparts.

[1]https://archive.ics.uci.edu/

[2]https://github.com/sdv-dev/CTGAN

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C. Baselines and evaluation metrics

We compared the proposed method with two recent SOTA DP methods, GCkM-DP [42] and OC-DP [43], on the three datasets. We resolved critical limitations in these baselines; therefore, they are ideal choices for comparison. The utility of T_P was evaluated using three metrics. We analyzed the privacy guarantees theoretically and verified them via three attack scenarios: data reconstruction, SA disclosure, and SA prediction/inference. We identified three crucial tasks to assess the utility of our method. Specifically, we analyzed and compared our method with existing methods using information loss ($ILoss$), classification error rate (CE_{rate}), and query error rate (QE_{rate}). $ILoss$ was measured via mean absolute error (MAE) and total variational distance (TVD) for X_{nu} and X_{ca}, respectively:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|\zeta_i - \hat{\zeta}_i\right| \tag{20}$$

where ζ_i and $\hat{\zeta}_i$ are true and noisy values, and

$$\text{TVD}(\eta, \eta^{'}) = \frac{1}{2}\sum_{x\in\Omega}\left|\eta(x) - \eta^{'}(x)\right| \tag{21}$$

where η and $\eta^{'}$ denote the real and noisy distributions over the same domain Ω.

The CE_{rate} of T_P is measured using Eq. 22:

$$CE_{rate} = 1 - Accuracy = 1 - \frac{T_p + T_n}{|T_P|} \tag{22}$$

In Eq. 22, T_p and T_n indicate true positive and true negative, respectively, and $|T_P|$ is the total number of records in T_P.

The QE_{rate} for T_P is measured using Eq. 23:

$$QE_{rate} = \frac{|R_T - R_{T_P}|}{R_T} \tag{23}$$

where R_T is the result from T, and R_{T_P} is the result from T_P.

The reason to use three diverse metrics is to prove the generalizability of the results from ML utility as well as from a general perspective (e.g., data mining). Apart from the above metrics, we evaluated the efficacy of our method in terms of PP against three attack scenarios: probabilistic SA disclosure (SAD), attacker accuracy (AA) in predicting an SA, and data reconstruction (DRC), as formalized in equations 24 to 26:

$$SAD = Pr[X_{iy} = X_{iy}^{true}|T_P, \varrho] \tag{24}$$

where X_{iy} is any random value related to the SA of the ith individual, X_{iy}^{true} is the actual SA of the ith individual, T_P is the noisy dataset, and ϱ denotes the attacker's knowledge (in the worst-case scenario);

$$AA = \frac{1}{N}\sum_{i=1}^{N}\mathbf{1}[\hat{y}_i = y_i] \tag{25}$$

where $1[.]$ is an indicator function for correct matches, $\hat{y}_i$ and y_i are the predicted and the true SA of the ith individual;

$$DRC_{a,b}^{J} = \frac{\#\{i : c_i = a \text{ and } \hat{c}_i = b\}}{\#\{i : c_i = a\}} \tag{26}$$

where a and b are the true and predicted labels in a particular column, c_i is the true class of record i, and $\hat{c}_i$ is the class predicted by the attacker for record i.

Using the above seven metrics, privacy and utility results were computed and contrasted with the baselines.

D. Empirical results and comparisons

1) Utility results and comparisons: In the first set of experiments, we computed $ILoss$ using Eqs. 20 and 21 and compared results with two DP methods. Tables VII and VIII present the overall MAE and TVD results computed from each T.

Table VII
AVG. MAE: PROPOSED METHOD VERSUS RECENT SOTA METHODS.

T	Avg. $ILoss(MAE)$		
	Ours	GCkM-DP [42]	OC-DP [43]
Adults (Synthetic)	46.83	135.15	102. 94
COMPAS	26.42	75.89	32.54
Bank Marketing	16,064.51	38,468.81	40,036.73

Table VIII
AVG. TVD: PROPOSED METHOD VERSUS RECENT SOTA METHODS.

T	Avg. $ILoss(TVD)$		
	Ours	GCkM-DP [42]	OC-DP [43]
Adults (Synthetic)	1.26	2.92	2.95
COMPAS	0.29	0.39	1.94
Bank Marketing	1.01	3.46	3.63

From the results, we can see that the COMPAS dataset has a lower $ILoss$ because of its lower dimensionality, whereas the Bank Marketing dataset has the highest $ILoss$ because of its higher dimensionality. In the MAE metric, GCkM-DP [42] yielded poor performance because noise is added twice (values are replaced with cluster centers followed by the LM application). In contrast, the TVD of OC-DP [43] was poor because the combinations were not created before EM application. In the proposed method, noise is added based on the records' sensitivity, and combinations are created before EM application, resulting in better performance across both metrics. The proposed method significantly outperformed the existing methods across all three datasets for both MAE and TVD. It is worth noting that the ϵ value has a significant impact on the results.

To show the impact of ϵ on utility, we present in Fig. 9 a breakdown of MAE results for each method on the Adults dataset. The number of ϵ values used in each technique were three, seven, and one for our method, OC-DP, and GCkM-DP, respectively. In our method and OC-DP, the optimal ϵ is determined for each partition or cluster, whereas GCkM-DP uses a fixed value for ϵ. For a fair assessment, we took the average of the three smaller ϵ values (0.01, 0.1, and 1) used in the GCkM-DP method to add noise to the data. The reason for using a smaller value is to assess the impact more accurately,

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since ϵ greater than 1 often does not yield a detailed analysis of the privacy-utility trade-off. Additionally, this method involves two changes in X_{nu} (replacing the real value with the cluster center, and adding noise), as indicated by changes 1 and 2 in Fig. 9(c). The values for ϵ in our method were 0.39, 0.44, and 0.72 for p0, p1, and p2, respectively. The values for ϵ in OC-DP were 0.168, 0.164, 0.10, 0.11, 0.26, 0.125, and 0.106 for C0 to C6, respectively.

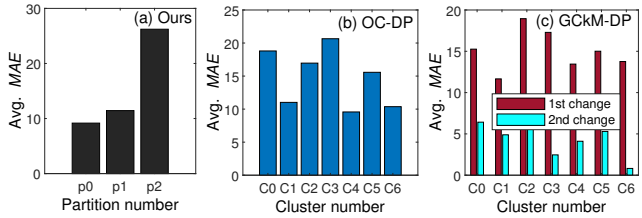

Figure 9. Comparisons of MAE against the SOTA baseline methods.

From the results, we can see that the proposed method makes fewer modifications in the data than its counterparts. These results demonstrate the superiority of our method in terms of utility (i.e., lower MAE).

In the second set of experiments, we trained an ML SVM classifier on T_P generated by each method and compared the resulting CE_{rate} values. Specifically, we trained the classifier with different splits of training and testing data and computed the average CE_{rate} for comparative analysis. The results are shown in Fig. 10, where we present the average and best results for each method. For better analysis, we also report the CE_{rate} from non-perturbed data (T).

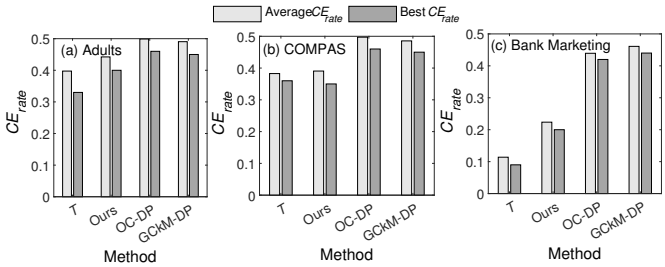

Figure 10. CE_{rate} comparisons of our method against the SOTA baselines.

As shown in Fig. 10, the CE_{rate} for T was low with no perturbations. Among the perturbation algorithms, our method yielded a lower CE_{rate} due to fewer modifications in T_P, and the results are very close to T. These results fortify the efficacy of our method from the perspective of ML utility.

In the third set of experiments, we ran various queries on T_P and computed QE_{rate} using Eq. 23. To evaluate utility, we executed queries on both X_{nu} and X_{ca} in forms such as the sum of a column, the mean of a column, the least and most frequent values, and the count of individuals with a particular value. Fig. 11 presents results for the overall QE_{rate} from all three datasets. We can see that our method yielded correct answers for most queries across all datasets. These results emphasize the significance of the proposed method from a data mining perspective, where query answers with sufficient accuracy are desirable.

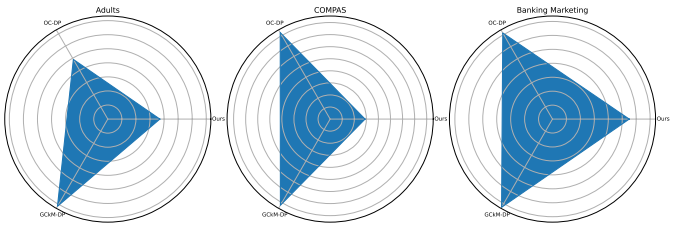

Figure 11. QE_{rate} comparisons of our method against the SOTA baselines.

To further validate the effectiveness of the proposed method, we separately computed the QE_{rate} for X_{ca} (categorical only), X_{nu} (numerical only), and $X_{ca}+X_{nu}$ (mixed data). Fig. 12 presents the QE_{rate} results for the three different types of query. GCkM-DP [42] and OC-DP [43] add too much noise, leading to poor QE_{rate} results. In our method, less noise is injected via the optimal ϵ; thus, all three query types can be answered with sufficient accuracy across datasets.

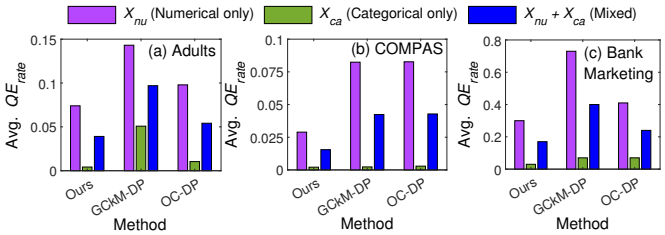

Figure 12. QE_{rate} comparisons of our method for different types of query.

Based on the detailed experiments and comparisons using four metrics, our method yielded higher utility than the SOTA DP methods used for comparison. The data produced by our method can significantly contribute to secondary use.

2) Privacy results and comparisons: In the fourth set of experiments, we empirically evaluated the proposed method in terms of PP against three different types of privacy attack: PP robustness in terms of posterior probability (gain in an attacker's guidance/knowledge about a target individual after seeing T_P), the prediction/estimation of sensitive information from T_P, and generating a true copy of T from T_P by using auxiliary information. All three threats are practical and difficult to defend against. We assumed worst-case scenarios when computing and comparing privacy disclosures (e.g., the attacker might possess background knowledge of random records in the data; the attacker might try to reconstruct a column; or the attacker might try to predict an individual's SA). We computed the privacy risks using Eqs. 24 to 26, and the corresponding results are in Table IX.

In the first attack, we computed the risk of SA disclosure by exposing certain parts of a QID to the attacker. In the second attack, we showed different QIDs (first a few and then all of them) to the attacker and forced the attacker to predict the SA of the target. In the second block, we present the results for both settings. From this, we can see that when entire columns are provided, it becomes difficult for the attacker to predict

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Table IX
PP AGAINST THREE ATTACKS: OUR METHOD VS. BASELINE METHODS.

Attack Type	T	Ours	GCkM-DP [42]	OC-DP [43]
SADR	Adults (Synthetic)	0.070	0.068	0.052
	COMPAS	0.193	0.329	0.348
	Bank Marketing	0.181	0.174	0.190
AA	Adults (Synthetic)	0.122 → 0.018	0.118 → 0.017	0.087 → 0.018
	COMPAS	0.328 → 0.057	0.397 → 0.261	0.414 → 0.282
	Bank Marketing	0.325 → 0.036	0.290 → 0.058	0.328 → 0.052
DRC	Adults (Synthetic)	0.264	0.263	0.269
	COMPAS	0.338	0.348	0.349
	Bank Marketing	0.250	0.291	0.252

the SA correctly. In the third attack, we allow the attacker to reconstruct the whole, randomly chosen, categorical QID column. In all these settings, our method achieved competitive performance relative to the baseline methods across most tests. These results prove that utility improvements can be achieved without compromising privacy. Detailed ablation experiments to demonstrate the superiority of our method under various conditions and method choices are presented in **Appendix B**.

Impact of large number of clusters on the performance: To further verify the robustness of our method for more clusters similar to GCkM-DP [42], we expanded the three partitions into more clusters like GCkM-DP [42], where the number of clusters and data points (or records) are same as of GCkM-DP baseline. For each dataset, we determine the optimal number of clusters using the elbow method, then divide the records (i.e., data points) into those clusters. Afterward, we analyzed the privacy and utility results in each cluster and contrasted them with the GCkM-DP baseline. In the first test, we quantify the utility results and contrast them with the GCkM-DP [42] baseline, and the corresponding results are in Table X.

Table X
AVG. MAE AND TVD PER CLUSTER: PROPOSED METHOD VERSUS GCkM-DP BASELINE.

T	Cluster	Avg. $ILoss(MAE)$		Avg. $ILoss(TVD)$	
		Ours	GCkM-DP [42]	Ours	GCkM-DP [42]
Adults (Synthetic)	C0	2.90	21.68	0.13	0.43
	C1	3.05	16.55	0.12	0.43
	C2	3.22	24.72	0.17	0.42
	C3	5.50	19.75	0.19	0.42
	C4	6.94	17.56	0.22	0.43
	C5	13.50	20.31	0.42	0.41
	C6	11.72	14.58	0.43	0.42
COMPAS	C0	13.41	20.67	0.095	0.41
	C1	2.01	14.9	0.091	0.38
	C2	2.95	16.97	0.11	0.41
	C3	4.21	12.9	0.21	0.39
	C4	3.85	9.81	0.37	0.36
Bank Marketing	C0	171.05	230.55	0.13	0.41
	C1	315.21	306.33	0.11	0.38
	C2	398.19	538.98	0.12	0.40
	C3	390.25	378.63	0.21	0.39
	C4	475.01	1029.19	0.23	0.36
	C5	597.09	1111.69	0.30	0.32
	C6	11113.31	14822.21	0.28	0.33
	C7	1198.13	7320.71	0.29	0.36
	C8	1105.48	1985.57	0.26	0.32
	C9	10303.53	10312.91	0.32	0.35

Referring to Table X, it can be observed that $ILoss$ is different for each cluster due to different sizes. In the comparative analysis, our method significantly outperformed the baseline on $ILoss$, as quantified by MAE and TVD. Across 44 tests, our method yielded only five instances/clusters in which it could not outperform the baseline on either MAE or TVD. However, it is worth noting that most of those clusters are part of ρ_h, and require higher PP. In contrast, the GCkM-DP yielded higher $ILoss$, particularly in MAE, as noise is added twice (replacement of numerical data with cluster centers followed by noise addition). These results underscore the efficacy of the proposed method, which consistently yields higher utility than the baseline across clusters. Next, we analyze and compare the performance of our method with GCkM-DP across clusters from the PP perspective using the DRC and $SADR$ metrics, and present the results in Fig. 13.

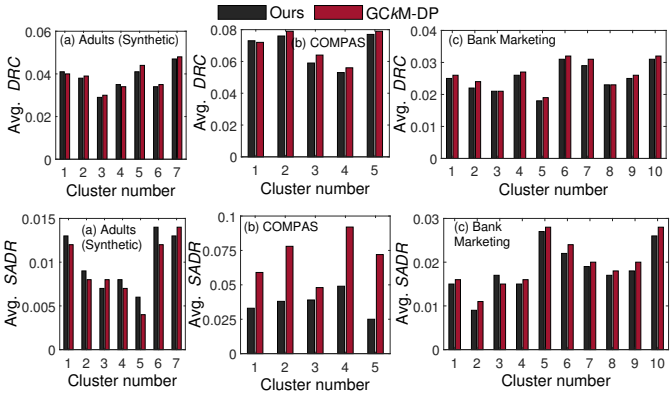

Figure 13. PP comparisons of our method with the GCkM-DP across clusters.

Referring to Fig. 13, it can be observed that the privacy leakage for the two metrics differs across clusters due to differences in record counts and data composition. Our method has provided either similar or better privacy guarantees across clusters in most datasets, though not in all cases. The relatively higher $SADR$ in the adults dataset was due to a higher imbalance in SA values (e.g., one value was more dominant than the others). Further, the clusters with relatively higher privacy leakage result from either ρ_m/ρ_l, where rigorous PP is not required. In contrast, the GCkM-DP method does not account for sensitivity and applies the same PP across clusters, which can lead to inconsistent privacy guarantees (i.e., over-protection of less sensitive data). Based on the experimental findings and analysis, the following choices can be made in realistic scenarios regarding PP methods.

1) GCkM-DP [42] is a better choice when either all data is sensitive or higher PP is desirable at the expense of data utility.
2) Proposed method is a better choice when either not all data is sensitive or a greater balance between privacy and utility is desirable.
3) Proposed method is a better choice when personalized privacy requirements need to be met, or perturbed data is intended to be used in personalized/data-driven services.

The application of the proposed method to larger (i.e., more records and attributes) datasets, along with the corresponding results and comparisons, is given in **Appendix C**. Feature-based comparisons of studies centering evolutionary optimization and cluster-level DP methods are provided in **Appendix D.** Lastly, the contribution of our method to privacy-preserving analytics is discussed in **Appendix E.**

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VII. Conclusion

In this work, we propose a novel ϵ-DP-based method for structured data perturbation that optimizes the privacy-utility trade-off, facilitating secondary use of the data. We introduce three novel optimizations to the DP model to reduce the need for heavier modifications, thereby improving data utility in downstream tasks without compromising privacy guarantees. The first optimization provides an algorithmic solution to classify data by sensitivity, eliminating the need for user input because, in most cases, users are unable to determine their data's sensitivity. The second optimization contributes to determining the optimal ϵ by balancing both privacy and utility, whereas existing methods only consider privacy when determining ϵ via PSO, resulting in poor data utility. The last optimization helps preserve the essential statistical characteristics of the raw data in the perturbed data, thereby enhancing the benefits of DS. Using seven diverse evaluation metrics, extensive experiments on real benchmark and synthetic datasets demonstrate the superiority of our method over recent SOTA DP methods in terms of privacy preservation and utility enhancement. Theoretical analysis proved that our method is ϵ-DP. In the future, we intend to apply our method to datasets with multiple SAs rather than a single SA. Finally, we want to optimize our results and make our method lightweight by integrating user feedback into the sensitivity analysis.

Acknowledgments

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